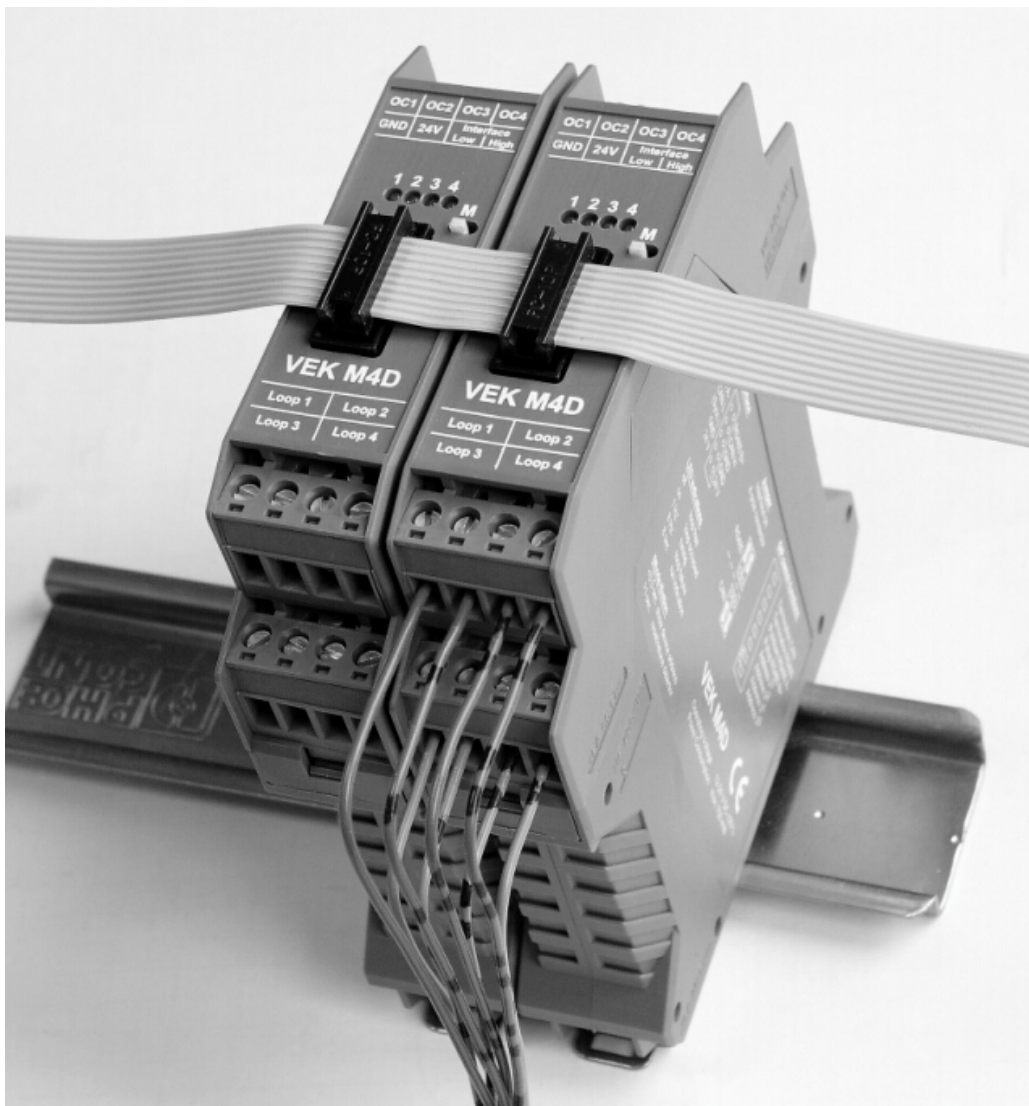


CANopen Interface Object Dictionary

Traffic Detector - VЕК M4D



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1 General

The VEK M4D traffic detector is compliant with the CANopen Standard 301 and DI401. The specifications contained in the following chapters describes the manufacturer specific objects of the object dictionary (see CiA 301). Furthermore there are four mandatory objects at the indices 1000h, 1001h, 1017h, 1018h, six device profile DI401 objects and six optional object at the indices 1005h, 1008h, 1009h 100Ah, 1014h and 1800h supported by VEK M4D.

For more information about CANopen refer to CiA.

Transmission rates: 20, 50, 100, 125, 250, 500, 800, 1000 kbps

Factory default setting: 250 kbps

1.1 Reset

If the NMT Master sends a **Reset Node Protocol** (COB-ID=0 and NMT command specifier **cs**=129) to a detector, then the detector is reset.

If the NMT Master sends a **Reset Communication Protocol** (COB-ID=0 and NMT command specifier **cs**=130) to a detector, then only the CAN hardware of the detector is reset. All data in RAM it will be remain valid. After both Reset Protocols the detector will send a Boot-Up Message (COB-ID=1792+detector address with one data byte with value 0) and it will enter autonomously the PRE-OPERATIONAL state.

2 CiA Draft Standard 301 Mandatory Objects

2.1 Device Type 1000h

Data Type	Unsigned32	
Access Type	read only	
Value Description	00000000h	device type

2.2 Error Register 1001h

Data Type	Unsigned8	
Access Type	read only	
Value Description	00h	generic error

2.3 Identity Object 1018h

2.3.1 Vendor ID 1018sub1

Data Type	Unsigned32	
Access Type	read only	
Value Description	000001FEh	unique value allocated to FEIG ELECTRONIC GmbH

2.3.2 Product Code 1018sub2

Data Type	Unsigned32	
Access Type	read only	
Value Description	00000001h	

2.3.3 Revision Number 1018sub3

Data Type	Unsigned32	
Access Type	read only	
Value Description	e.g. 02600300h	Bit31-16 represents the major revision number Bit15-0 represents the minor revision number

2.3.4 Serial Number 1018sub4

Data Type	Unsigned32	
Access Type	read only	
Value Description	00000001h	

3 CiA Draft Standard 301 Optional Objects

3.1 COB-ID SYNC Message 1005h

Index 1005h defines the COB-ID of the Synchronisation Object (SYNC). Only 11-bit ID (CAN 2.0A) are supported (Bit 29 is read only). The detector is not able to generate SYNC messages (Bit 30 is read only).

Data Type	Unsigned32	
Access Type	read / write	
Value Description	00000080h	(default)
Bit 31 (MSB)	X	don't care
Bit 30 (read only)	0	Device does not generate SYNC message. (default)
	1	Device generates SYNC message.
Bit 29 (read only)	0	11-bit ID (CAN 2.0A) (default)
	1	29-bit ID (CAN 2.0B)
Bit 28-11	0	if Bit 29 = 0
	X	if Bit 29 = 1: Bit 28-11 of 29-bit-SYNC-COB-ID
Bit 10-0 (LSB)	X	Bits 10-0 of SYNC-COB-ID

3.2 Manufacturer Device Name 1008h

Data Type	Visible String (ASCII)	
Access Type	const	
Value Description	"VEK M4D"	

3.3 Manufacturer Hardware Version 1009h

Data Type	Visible String (ASCII)	
Access Type	const	
Value Description	e.g "1.00"	

3.4 Manufacturer Software Version 100Ah

Data Type	Visible String (ASCII)	
Access Type	const	
Value Description	e.g " M4D-A V3.00 27.06.06"	

3.5 COB-ID Emergency Object 1014h

Index 1014h defines the COB-ID of the Emergency Object (EMCY). Only 11-bit ID (CAN 2.0A) is supported (Bit 29 is read only).

Data Type	Unsigned32	
Access Type	read / write	
Value Description	00000080h + Node-ID	(default)
Bit 31-30	0	reserved
Bit 29 (read only)	0	11-bit ID (CAN 2.0A) (default)
	1	29-bit ID (CAN 2.0B)
Bit 28-11	0	if Bit 29 = 0
	X	if Bit 29 = 1: Bit 28-11 of 29-bit-EMCY-COB-ID
Bit 10-0 (LSB)	X	Bits 10-0 of EMCY-COB-ID

3.6 Producer Heartbeat Time 1017h

Data Type	Unsigned16	
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Access Type	read/write	
Value Description	0000h	(default) heartbeat time is not used
	0001h...FFFFh (milliseconds)	cycle time of the heartbeat

3.7 TPDO1 Communication Parameter 1800h

3.7.1 COB-ID used by PDO 1800sub1

Data Type	Unsigned32	
Access Type	read / write	
Value Description	180h + Node-ID	(default)
Bit 31 (MSB)	0	PDO valid
	1	PDO not valid
Bit 30 (read only)	0	RTR allowed on this PDO
	1	no RTR allowed on this PDO (default)
Bit 29 (read only)	0	11-bit ID (CAN 2.0A) (default)
	1	29-bit ID (CAN 2.0B)
Bit 28-11	0	if Bit 29 = 0
	X	if Bit 29 = 1: Bit 28-11 of 29-bit-EMCY-COB-ID
Bit 10-0 (LSB)	X	Bits 10-0 of COB-ID

3.7.2 Transmission Type 1800sub2

Data Type	Unsigned8	
Access Type	read / write	
Value Description	255	(default)
	0	acyclic, synchronous
	1...240	cyclic, synchronous
	241...251	reserved
	252	synchronous, RTR only (not supported)
	253	asynchronous, RTR only (not supported)
	254	asynchronous, if PDO changes its value it will be transmitted immediately
	255	asynchronous, if PDO changes its value it will be transmitted immediately

3.7.3 Inhibit Time 1800sub3

Data Type	Unsigned16	
Access Type	read / write	
Value Description	0	not used (default)
	0001h...FFFFh (1/10 millisecond)	minimum interval for PDO transmission

If during *inhibit time* interval a status change on the digital inputs (see 5.1) occurs, these changes will be send with the next TPDO.

3.7.4 Event Timer 1800sub5

Data Type	Unsigned16	
Access Type	read / write	
Value Description	0	not used (default)
	0001h...FFFFh (millisecond)	time to elapse to generate an event

4 CiA Draft Standard 301 Codes

4.1 Abort SDO Codes

The table below contains a part of the abort codes defined by CiA that the detector will send if an SDO error encounters.

Abort code	Description
0601 0002h	Attempt to write to a read only object
0601 0001h	Attempt to write to a write only object
0604 0043h	General parameter incompatibility reason.
0607 0010h	Data type does not match, length of service parameter does not match.
0609 0011h	Sub-Index does not exist.
0609 0030h	Value range of parameter exceeded.
0609 0031h	Value of parameter written too high.
0609 0032h	Value of parameter written too low.

The other defined abort codes are not used.

4.2 Emergency Error Codes

The table below contains a part of the emergency error codes defined by CiA that the detector will send with an emergency object (EMCY) if a device internal error situation occurs.

Error code	Description
0000	Error Reset or No Error
8140	Recover from bus off
8100	CAN in Warning Mode(error active) (Receive or transmit error counter > 96)
8110	CAN Overrun (Object Lost)
8120	CAN in Error Passive Mode
8150	CAN Transmit CAN Object Collision (Received CAN Message has a Transmit COB-ID)
8130	Heartbeat Error (only if Heartbeat Consumer is activated)

The other defined error codes are not used.

5 CiA Draft Standard 401 Objects

There are six device profile specific objects: 6000h, 6002h, 6005h...6008h supported by VEK M4D. The object 6002h can be used to change the polarity of the inputs. The object 6003h (Filter Constant) aren't supported; instead of them the object 2FC1sub2 (see 10.2.2) can be used to activate a manufacturer specific filter. The objects 6005h up to 6008h can be used to change the transmission behavior of the TPDO1. For more information about these objects refer to CiA Standard 401.

5.1 Read Input 8Bit 6000h

The traffic detector VEK M4D supports 16 digital inputs that are described in the following chapters.

5.1.1 Read Input 8Bit 6000sub1

Data Type	Unsigned8	
Access Type	read only	
Bit0	0	Loop 1 is not covered
	1	Loop 1 is covered
Bit1	0	Loop 2 is not covered
	1	Loop 2 is covered
Bit2	0	Loop 3 is not covered
	1	Loop 3 is covered
Bit3	0	Loop 4 is not covered
	1	Loop 4 is covered
Bit4	0	Loop 1 has no fault
	1	Loop 1 fault
Bit5	0	Loop 2 has no fault
	1	Loop 2 fault
Bit6	0	Loop 3 has no fault
	1	Loop 3 fault
Bit7	0	Loop 4 has no fault
	1	Loop 4 fault

5.1.2 Read Input 8Bit 6000sub2

Data Type	Unsigned8	
Access Type	read only	
Bit0	0	Loop 1 is not adjusting
	1	Loop 1 is adjusting
Bit1	0	Loop 2 is not adjusting
	1	Loop 2 is adjusting
Bit2	0	Loop 3 is not adjusting
	1	Loop 3 is adjusting
Bit3	0	Loop 4 is not adjusting
	1	Loop 4 is adjusting
Bit4	0	Direction A not detected by Module 1
	1	Direction A detected by Module 1
Bit5	0	Direction B not detected by Module 1
	1	Direction B detected by Module 1
Bit6	0	Direction A not detected by Module 2
	1	Direction A detected by Module 2
Bit7	0	Direction B not detected by Module 2
	1	Direction B detected by Module 2

6 Loops 2000h-227Fh

6.1 Loop Parameter 200xh

Parameters to configure the loops. Letter „x“ has to be replaced by 1, 2, 3 or 4 for the loop.
If the values of loop parameters are changed, the new values are stored immediately to EEPROM.

6.1.1 Frequency Range/Step 200xsub1

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h	automatic, depend on address
	01h	range 30-40Hz
	02h	range 45-55Hz
	03h	range 60-75Hz
	04h	range 80-100Hz (default)
	05h	range 105-140Hz
	06h...10h	reserved
	11h	transformer step 1
	:	:
	18h	transformer step 8
	19h..FFh	reserved

6.1.2 Sensitivity 200xsub2

Data Type	Unsigned8		
Access Type	read/write		
Value Description	0	Threshold 4	0,005% $\Delta f/f$
	1	Threshold 10	0,013% $\Delta f/f$
	:	:	
	12	Threshold 120 (def.)	0,150% $\Delta f/f$
	:	:	
	255	Threshold 2550	3,188% $\Delta f/f$

6.1.3 Hysteresis 200xsub3

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h...13n	reserved
	14h..50h	20...80% (default: 75%)
	51h...FFh	reserved

6.1.4 Hold Time 200xsub4

Data Type	Unsigned8	
Access Type	read/write	
Value Description	0: infinite	
	01h...FFh	minutes (default: 20 minutes)

6.2 Current Loop Values 208xh

6.2.1 Frequency 208xsub1

Data Type	Unsigned8	
Access Type	read only	
Value Description	0..255	Frequency [kHz]

If a request occurs immediately after frequency setting or reset, the answer contains 0 kHz. Please try again after end of retuning!

6.2.2 Amplitude 208xsub2

Data Type	Integer16	
Access Type	read only	
Value Description	-32767....32767	

6.2.3 Maximum Amplitude 208xsub3

Data Type	Integer16	
Access Type	read only	
Value Description	-32767....32767	

Maximum amplitude of last object!

6.3 Loop Status 210xh

6.3.1 Covered level sensitive 210xsub1

Data Type	Boolean	
Access Type	read/write	
Value Description	0	covered
	1	free

6.3.2 Covered low amplitude 210xsub2

Data Type	Boolean	
Access Type	read only	
Value Description	0	covered
	1	free

6.3.3 Adjusting 210xsub3

Data Type	Boolean	
Access Type	read only	
Value Description	0	Adjusting completed.
	1	Adjusting in progress.

6.3.4 Offline 210xsub4

Data Type	Boolean	
Access Type	read only	
Value Description	0	Loop is active
	1	Loop is offline

6.3.5 Fault Status 210xsub5

Data Type	Unsigned8	
Access Type	read only	
Value Description	00h	No fault
	01h	Loop fault
	02h	Frequency out of settings
	03h...FFh	reserved

6.3.6 Change Loop Status 210xsub6

Data Type	Boolean	
Access Type	write only	
Value Description	0	don't change Loop Status
	1	change Loop Status

Setting of loop status (210xsub6 is set) overwrites the evaluated loop status with the value contained in **Covered level sensitive** (210xsub1). It is only temporary and will not stored to EEPROM.

6.4 Timer 218xh

6.4.1 Start Time (minutes) 218xsub1

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	Time Base value at begin of loop covering

6.4.2 Stop Time (minutes) 218xsub2

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	Time Base value at end of loop covering

6.4.3 Busy Time (minutes) 218xsub3

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	minutes

If loop is free, busy time of last object is transmitted ! For a covered loop the actual busy time is transmitted.
No Overflow! If busy time is greater than FFFFh = 65535 the maximum value 65535 is transmitted!

6.4.4 Gap Time (minutes) 218xsub4

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	minutes

If loop is free, the actual gap time is transmitted. If loop is covered, the gap time before covering is transmitted.
No Overflow! If gap time is greater than FFFFh = 65535 the maximum value 65535 is transmitted!

6.4.5 Start Time (seconds) 218xsub5

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	Time Base value at begin of loop covering

6.4.6 Stop Time (seconds) 218xsub6

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	Time Base value at end of loop covering

6.4.7 Busy Time (seconds) 218xsub7

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	seconds

If loop is free, busy time of last object is transmitted ! For a covered loop the actual busy time is transmitted.
No Overflow! If busy time is greater than FFFFh = 65535 the maximum value 65535 is transmitted!

6.4.8 Gap Time (seconds) 218xsub8

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	seconds

If loop is free, the actual gap time is transmitted. If loop is covered, the gap time before covering is transmitted.
No Overflow! If gap time is greater than FFFFh = 65535 the maximum value 65535 is transmitted!

6.4.9 Start Time (milliseconds) 218xsub9

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	Time Base value at begin of loop covering

6.4.10 Stop Time (milliseconds) 218xsubA

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	Time Base value at end of loop covering

6.4.11 Busy Time (milliseconds) 218xsubB

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	milliseconds

If loop is free, busy time of last object is transmitted ! For a covered loop the actual busy time is transmitted.
No Overflow! If busy time is greater than FFFFh = 65535 the maximum value 65535 is transmitted!

6.4.12 Gap Time (milliseconds) 218xsubC

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	milliseconds

If loop is free, the actual gap time is transmitted. If loop is covered, the gap time before covering is transmitted.
No Overflow! If gap time is greater than FFFFh = 65535 the maximum value 65535 is transmitted!

6.5 Presence Signals 220xh

If the values of presence signal parameters are changed, the new values are stored immediately to EEPROM.

6.5.1 Output Channel 1 220xsub1

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 1 is not controlled by loop x
	1	output 1 is controlled by loop x

6.5.2 Output Channel 2 220xsub2

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 2 is not controlled by loop x
	1	output 2 is controlled by loop x

6.5.3 Output Channel 3 220xsub3

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 3 is not controlled by loop x
	1	output 3 is controlled by loop x

6.5.4 Output Channel 4 220xsub4

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 4 is not controlled by loop x
	1	output 4 is controlled by loop x

7 Direction Logic Module 2800h-28FFh

7.1 Direction Logic Parameter 280yh

Parameters to configure the direction logic module.

Letter „y“ has to be replaced by the 1 or 2 for the direction logic module.

If the values of direction logic parameters are changed, the new values are stored immediately to EEPROM.

7.1.1 Mode Direction A 280ysub1

Data Type	Unsigned8	
Access Type	read/write	
Value Description	0	logic D2
	1	logic D1
	2	logic DB
	3	logic F1 (default)
	4	logic F2
	5	logic FE
	6	logic SF
	7	logic BS
	8	logic PB

For definition of direction logic 0..8 see the detector description.

7.1.2 Mode Direction B 280ysub2

Data Type	Unsigned8	
Access Type	read/write	
Value Description	0	logic D2
	1	logic D1
	2	logic DB
	3	logic F1 (default)
	4	logic F2
	5	logic FE
	6	logic SF
	7	logic BS
	8	logic PB

For definition of direction logic 0..8 see the detector description.

7.1.3 Direction A detected 280ysub3

Data Type	Boolean	
Access Type	read only	
Value Description	0	not detected
	1	detected

7.1.4 Direction B detected 280ysub4

Data Type	Boolean	
Access Type	read only	
Value Description	0	not detected
	1	detected

7.1.5 Input Loop 1 280ysub5

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h	no input
	01h	loop1
	02h	loop2
	03h	loop3
	04h	loop4
	05h...FFh	reserved

7.1.6 Input Loop 2 280ysub6

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h	no input
	01h	loop1
	02h	loop2
	03h	loop3
	04h	loop4
	05h...FFh	reserved

7.2 Direction Signals 284yh

7.2.1 Output channel 1 direction A 284ysub1

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 1 is not controlled by direction logic module y when direction A is detected
	1	output 1 is controlled by direction logic module y when direction A is detected

7.2.2 Output channel 2 direction A 284ysub2

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 2 is not controlled by direction logic module y when direction A is detected
	1	output 2 is controlled by direction logic module y when direction A is detected

7.2.3 Output channel 3 direction A 284ysub3

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 3 is not controlled by direction logic module y when direction A is detected
	1	output 3 is controlled by direction logic module y when direction A is detected

7.2.4 Output channel 4 direction A 284ysub4

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 4 is not controlled by direction logic module y when direction A is detected
	1	output 4 is controlled by direction logic module y when direction A is detected

7.2.5 Output channel 1 direction B 284ysub5

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 1 is not controlled by direction logic module y when direction B is detected
	1	output 1 is controlled by direction logic module y when direction B is detected

7.2.6 Output channel 2 direction B 284ysub6

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 2 is not controlled by direction logic module y when direction B is detected
	1	output 2 is controlled by direction logic module y when direction B is detected

7.2.7 Output channel 3 direction B 284ysub7

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 3 is not controlled by direction logic module y when direction B is detected
	1	output 3 is controlled by direction logic module y when direction B is detected

7.2.8 Output channel 4 direction B 284ysub8

Data Type	Boolean	
Access Type	read/write	
Value Description	0	output 4 is not controlled by direction logic module y when direction B is detected
	1	output 4 is controlled by direction logic module y when direction B is detected

8 Output 2A00h-2BFFh

8.1 Output Parameter 2A0zh

Parameters to configure the outputs. Letter „z“ has to be replaced by the 1, 2, 3 or 4 for the output.
If the values of output parameters are changed, the new values are stored immediately to EEPROM.

8.1.1 Output Mode 2A0zsub1

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h	default off
	01h	default on
	02h	general fault mode
	03h	normal mode (default)
	04h	simulation mode
	05h..FFh	reserved

8.1.2 Invert Mode 2A0zsub2

Data Type	Boolean	
Access Type	read/write	
Value Description	0	non inverted signal (default)
	1	inverted signal

8.1.3 Error Flags 2A0zsub3

Data Type	Unsigned8	
Value Description (only for Output mode 03h)	00	Adjustment Flag is clear; Loop fault is clear
	01	Adjustment Flag is set; Loop fault is clear
	04	Adjustment Flag is clear; Loop fault is set (default)
	05	Adjustment Flag is set; Loop fault is set
	02h, 03h, 06h..FFh	reserved

8.1.4 Output On Delay Time 2A0zsub2

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h..FFh	z100ms (default : 0ms)

8.1.5 Output Off Delay Time 2A0zsub3

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h..FFh	x100ms (default : 0ms)

8.1.6 Output Min Signal Duration 2A0zsub4

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h	real signal output, no limitation
	01h...FFh	x100ms (default: 200ms)

8.1.7 Hardware Output Signal 2A0zsub5

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h	output channel z is manually deactivated
	01h	output channel z is manually activated
	02h	output channel z is automatically controlled (default)
	03h..FFh	reserved

Setting of hardware output signals (values 00h and 01h) overwrites the evaluated open collector signals according to the settings in *Output Mode*. Setting data is only temporary and will not stored to EEPROM! Use this protocol instead of 8.1.1 to set hardware output for devices with high operating cycles (e.g. flashing lights) to prevent EEPROM from damage.

9 Counter 2C00h-2C7Fh

For all counters overflow is possible! To reset the counters see 10.1 DeviceCommand 2FC0h.

9.1 Presence Counter 2C01h

9.1.1 Counter for loop 1

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	

9.1.2 Counter for loop 2

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	

9.1.3 Counter for loop 3

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	

9.1.4 Counter for loop 4

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	

9.2 Direction A Counter 2C02h

9.2.1 Direction Logic module 1 2C02sub1

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	

9.2.2 Direction Logic module 2 2C02sub2

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	

9.3 Direction B Counter 2C03h

9.3.1 Direction Logic module 1 2C03sub1

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	

9.3.2 Direction Logic module 2 2C03sub2

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	

9.4 Direction Parallel Traffic Counter 2C04h

9.4.1 Direction A 2C04sub1

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	

9.4.2 Direction B 2C04sub2

Data Type	Unsigned16	
Access Type	read only	
Value Description	0...65535	

10 Device 2FC0h-2FCFh

If the values of device parameter objects are changed, the new values are stored immediately to EEPROM. Only device command (2FC0h) object aren't stored to EEPROM.

10.1 DeviceCommand 2FC0h

Data Type	Unsigned32	
Access Type	write only	
Value Description	01h	Reset counter. All detector internal counter are set to zero.
	02h	All detector parameter in EEPROM and RAM data are set to their factory values. Note: CAN baudrate is reset to the default value of 500kbps.
	00h, 03h...FFh	reserved

10.2 MultiplexMode 2FC1h

10.2.1 Sampling Speed 2FC1sub1

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h	x1 (all loops) (default)
	01h	x2 (two loops)
	02h	x4 (one loop)
	03h...FFh	reserved

To assure a valid detector configuration *Sampling Speed* changes to value 00h, 01h or 02h causes automatic *MuxOrder* (see 10.2.3) changes to value E4h, 44h respectively 00h !!!

10.2.2 Filter 2FC1sub2

Data Type	Boolean	
Access Type	read/write	
Value Description	0	disabled (high speed)
	1	enable (high immunity) (default)

10.2.3 MuxOrder 2FC1sub3

Data Type	Unsigned8						
Access Type	read/write						
Value description			Time frame 1	Time frame 2	Time frame 3	Time frame 4	
			Bit1-0	Bit3-2	Bit5-4	Bit7-6	
			Corresponding bit group				
			Corresponding bit value				
	Loop 1	00b	00b				
	Loop 2	01b		01b			
	Loop 3	10b			10b		
	Loop 4	11b					11b

Each bit value describes a loop number that can be assigned to a time frame. All time frames must have assigned a bit value. After that, a byte value results if the bit groups of the time frames are correctly combined. The byte value for the example above is:

Byte value	Bit7-6	Bit5-4	Bit3-2	Bit1-0
11 10 01 00b (=E4h) (default value)	11b	10b	01b	00b

10.3 RS485Comm 2FC2h

10.3.1 Baudrate 2FC2sub1

Data Type	Unsigned8		
Access Type	read/write		
Value Description	01h	38400 Baud	
	02h	19200 Baud	
	03h	9600 Baud (default)	
	04h...FFh	reserved	

10.3.2 ParityEnable 2FC2sub2

Data Type	Boolean		
Access Type	read/write		
Value Description	0	even Parity	(default)
	1	odd Parity	

10.3.3 ParitySel 2FC2sub3

Data Type	Boolean		
Access Type	read/write		
Value Description	0	no Parity	
	1	Parity enable	(default)

10.4 RS485Protocol 2FC3h

10.4.1 ProtocolType 2FC3sub1

Data Type	Unsigned8	
Access Type	read/write	
Value Description	00h	this protocol (default)
	01h...FFh	reserved for future implementation

10.4.2 Test Mode 2FC3sub2

Data Type	Unsigned8	
Access Type	read/write	
Value Description	0	do not change, set always 0!

10.4.3 Address Offset & Hardware address 2FC3sub3

Data Type	Unsigned8	
Access Type	read (only)/write	
Value Description		
Bit 15-8	00h...0Fh*)	Hardware address: DIP Switch (read only)
Bit 7-0	00h...FEh**)	Address offset (default: 0)

*)for setting the address offset byte 0 must have the right DIP switch value

**)Range checking address offset: (setting DIP switch + address offset) < 255

The address offset remains unchanged unless the detector is not reset.

10.5 CANCom 2FC4h

10.5.1 Baudrate 2FC4sub1

Data Type	Unsigned8	
Access Type	read/write	
Value Description	1	20 kBaud
	2	50 kBaud
	3	100 kBaud
	4	125 kBaud
	5	250 kBaud (default)
	6	500 kBaud
	7	800 kBaud
	8	1000 kBaud

The CAN baud rate remains unchanged unless the detector is not reset.

10.6 Time base 2FC5h

10.6.1 TimeValue (minutes) 2FC5sub1

Data Type	Unsigned16	
Access Type	read only	
Value Description		0...65535 minutes

An overflow occurs after FFFFh = 65535.

65535 milliseconds	=> 65,5 seconds	
65535 seconds	=> 1092,25 minutes	=> 18,2 hours
65535 minutes	=> 1092,25 hours	=> 45,5 days

10.6.2 Time Value (seconds) 2FC5sub2

Data Type	Unsigned16	
Access Type	read only	
Value Description		0...65535 seconds

10.6.3 Time Value (milliseconds)

Data Type	Unsigned16	
Access Type	read only	
Value Description		0...65535 milliseconds

11 Process data objects

The detector VEK M4D supports the one Transmit PDO that are compliant with the CiA Standard 401. Receive PDOs are not supported. Also dynamic PDO mapping aren't supported.

11.1 Transmit PDO1 Mapping Parameter 1A00h

Mapped object			
Index	SubIndex	Description	Data length (bit)
6000	01	Read Input 8Bit 6000sub1	08h
6000	02	Read Input 8Bit 6000sub2	08h

For more information about the bit description of the Read Input objects see 5.1.

12 Notes

[illegible]